

SMB Underwriting Challenge

Understanding the Data & the End-to-End Logic

Intuit TechWeek NYC AI/ML Hackathon — strategy & data brief. Every number here is computed from the real dataset by `src/eda.py` and `src/report_pdf.py`.

1. The problem in one paragraph

You are a small-business lender. Using ~100K historical loan applications you must **(A)** decide whom to fund, **(B)** forecast how that funded book defaults over time, **(C)** answer causal “what-if” questions, and **(D)** defend it all in a 4-page-max writeup. It is deliberately **not** a single-metric Kaggle task: the data bakes in real underwriting pathologies (selection bias, censoring, informative missingness, confounding) and the score rewards economic reasoning, calibration, and causal honesty.

Every loan, if funded, has fixed terms: amount = requested, term = 60 days repaid by daily ACH draws, APR = 35%, origination fee = 3%. A loan defaults on 3 consecutive missed draws, 6 cumulative missed draws, or any balance left at day 90.

How the final score splits (drives where we spend time):

Component	Weight	Deliverable	What it measures
Portfolio P&L	0.30	A	Realized profit of the loans you funded
Trajectory	0.25	B	Accuracy of cohort default-timing curve
Calibration	0.20	A & B	Do your 90% intervals contain truth, not too wide
Causal	0.10	C	Closeness to true interventional effects
Writeup	0.15	D	Quality of methodological defense

Strategic read: A + B + calibration are 75% of the score and A & B share one model. C is only 10% — keep it simple and defensible; the points live in the §3 writeup defense.

2. The dataset at a glance

Frame	Rows	Cols	Notes
train	85,340	44	Outcomes only for prior-approved + matured loans
validation	4,489	44	Has outcomes — use to tune & calibrate
test	8,817	44	Outcomes withheld — you are scored on these

You decide on validation + test combined = **13,306** applicants (Deliverable A).

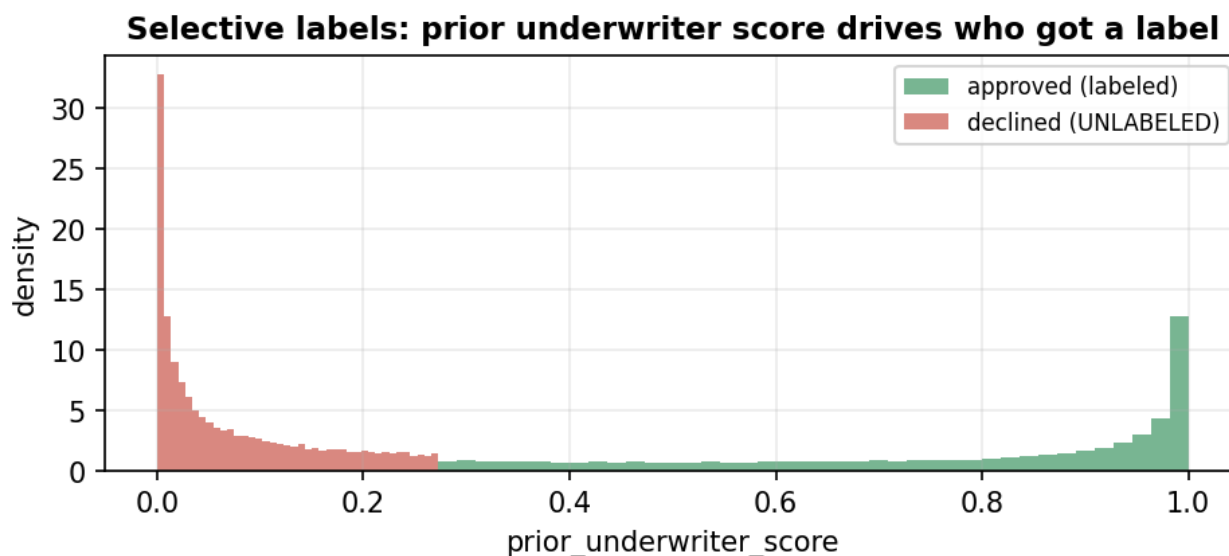
Features come in 8 groups: **business identity**, **self-reported** (applicant-stated), **bank-feed** (partial coverage), **bureau credit**, **platform engagement**, **application context**, **prior-underwriter** output, and **outcomes** (approved-only, must be dropped from inputs to avoid leakage).

3. Five things we must account for in the data

These are the assumption-violations that make the problem hard — and the writeup ammunition.

3.1 Selective labels / sample-selection bias

Repayment is observed **only** for the ~60.6% of loans a *prior* underwriter approved (51,722). The other 39.4% (33,618) were declined and are **unlabeled**. Train naively and you learn $P(\text{default} \mid \text{approved})$, not $P(\text{default} \mid \text{applied})$ — a biased view of the very applicants you must now newly decide on. The prior decision is near-deterministic in `prior_underwriter_score` (mean 0.781 approved vs 0.073 declined), so we model the propensity of approval and reweight (IPW) / report corrected vs uncorrected PD.

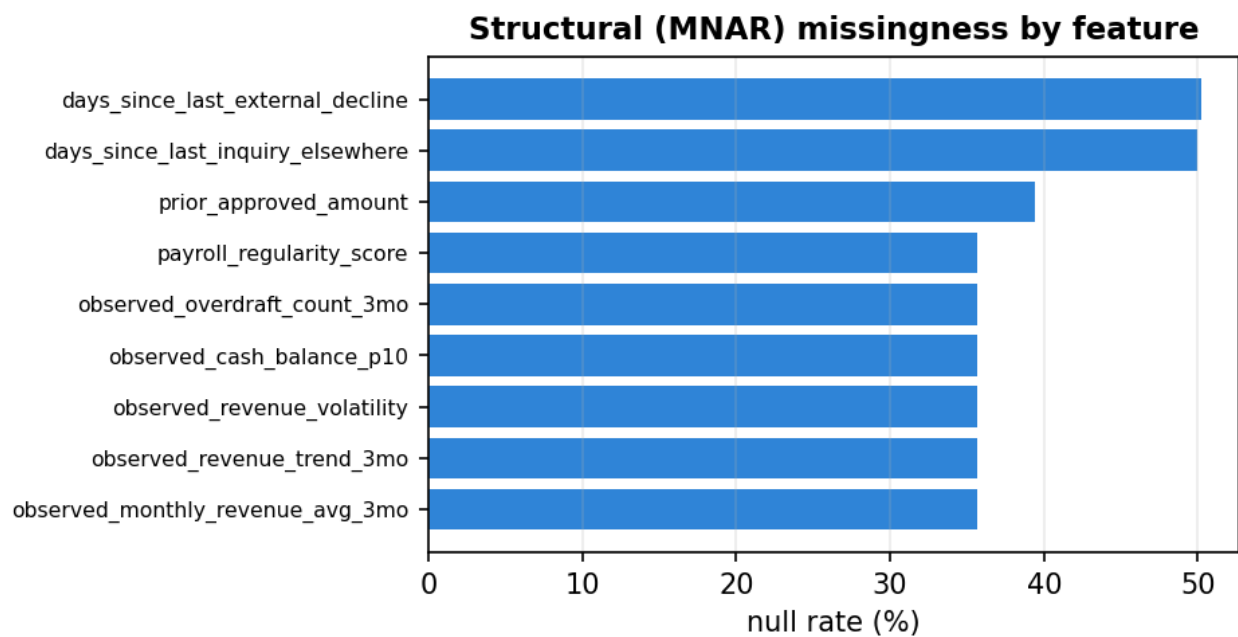


3.2 Censoring

Among labeled loans, 42,698 paid in full and 9,024 defaulted (**17.4%** default rate). Paid-in-full loans are *right-censored* at repayment (they simply never defaulted); immature/declined loans carry no event. This is a survival problem, not plain classification — we model time-to-default and treat repayment as censoring.

3.3 Informative (MNAR) missingness

Bank-feed columns are null exactly when no feed is linked (~35.7% of rows); “days since last decline/inquiry” is null when none occurred (~50%). The missingness is **not** random — its presence is itself a signal. We add explicit `*_is_missing` flags and sentinel-fill rather than imputing it away.



3.4 Optimistic self-report bias — but verify before claiming

Self-reported fields could be inflated vs. the bank-feed ground truth. In *this* data the effect is **mild**: stated/observed monthly revenue has median ~ 0.98 . Lesson: lean on observed bank-feed signals over stated ones, but don't overstate the bias in the writeup — the data says applicants are roughly honest on average.

3.5 Confounding (for the causal deliverable)

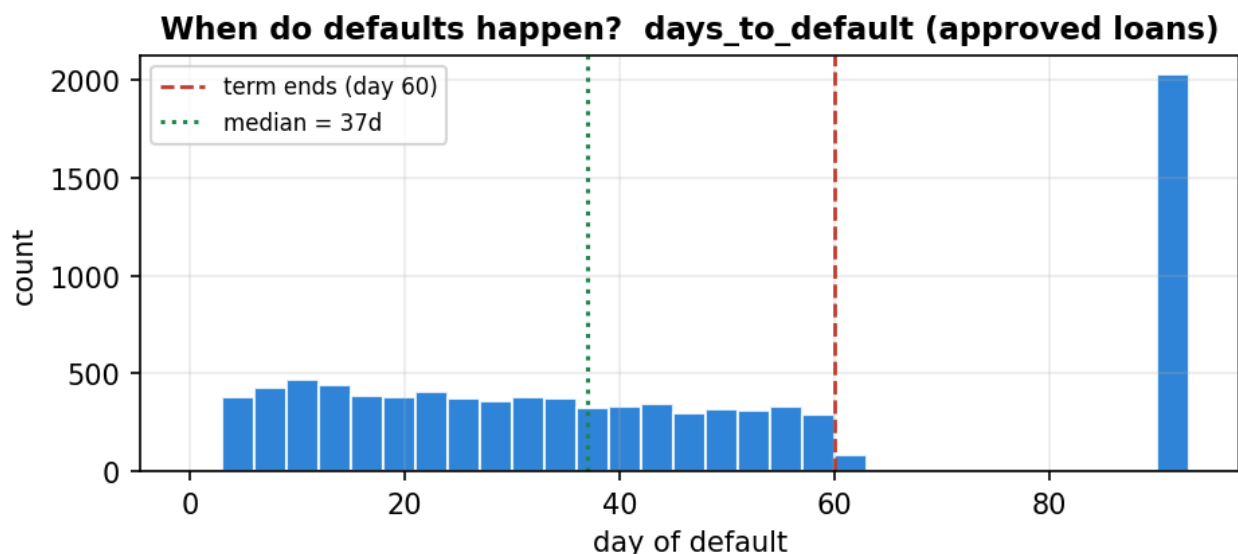
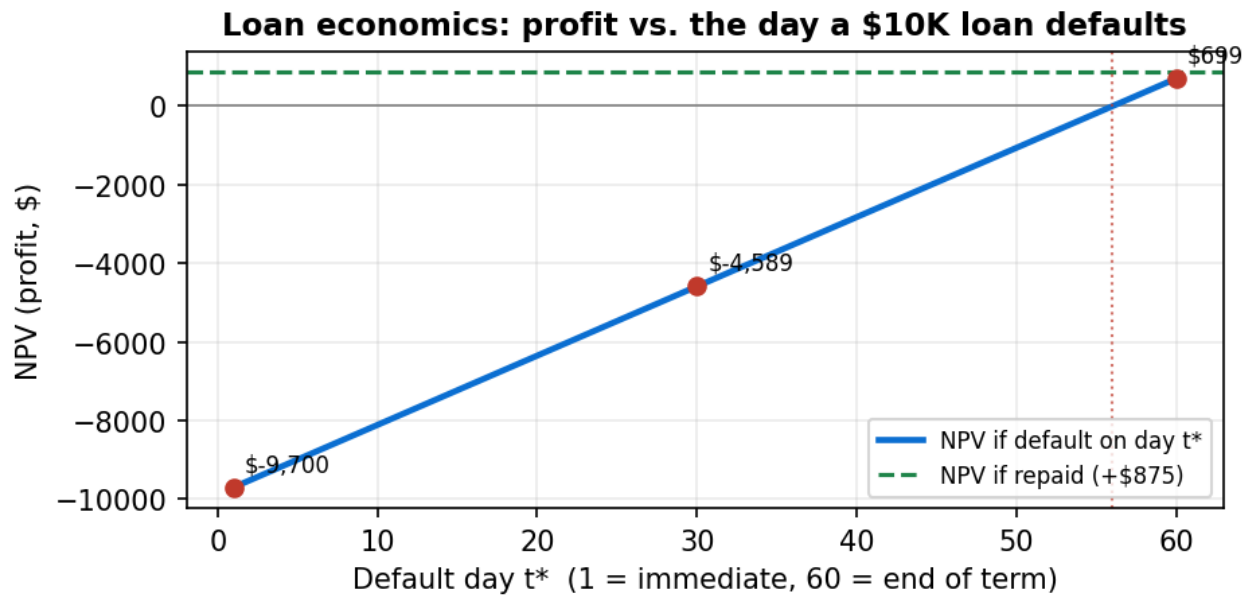
Deliverable C asks for $P(\text{default} \mid \text{do}(\text{feature} = v))$, not $P(\text{default} \mid \text{feature} = v)$. These differ whenever the feature is confounded (e.g. revenue drives both utilization and default). A naive “change the column and re-predict” answers the wrong (observational) question; the rubric explicitly rewards interventional reasoning.

4. The economics: why default *timing* drives everything

The correct funding decision is **not** a flat PD threshold — it is **approve** iff $E[NPV] > 0$. NPV depends on *when* a loan defaults, because the borrower pays a fixed daily draw until they stop:

- repaid: $NPV = F + R \cdot r \cdot (T/365)$
- default at day t^* : $NPV = F + D \cdot (t^* - 1) + \text{recovery} - R$
where $F = 0.03R$, $D = R(1 + rT/365)/T$, $r = 0.35$, $T = 60$.

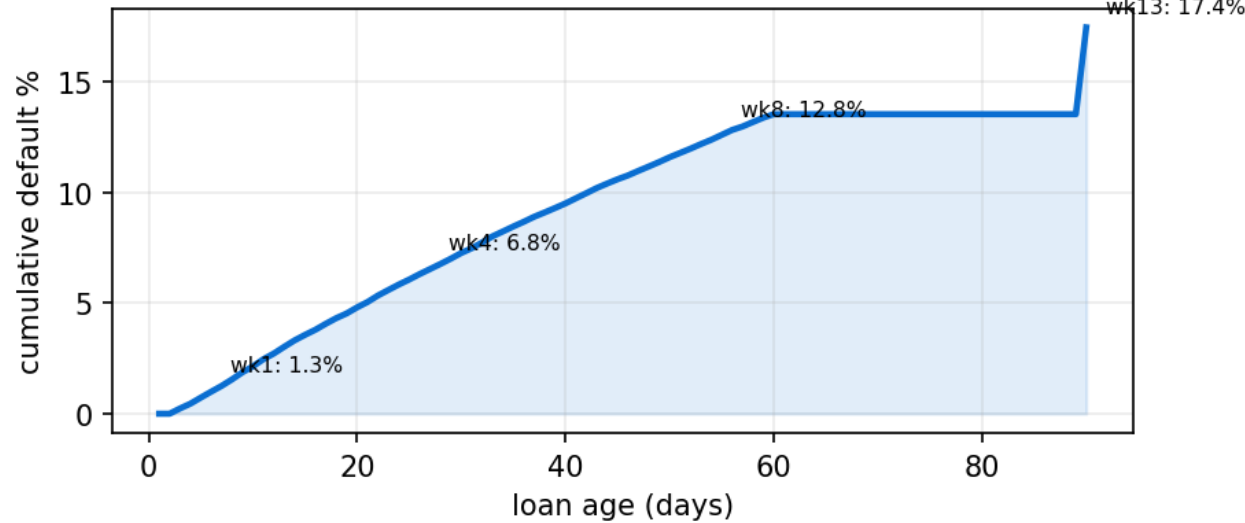
For a \$10K loan this means a repaid loan earns **+\$875**, a day-1 default loses **-\$9,700**, but a day-60 default is actually **+\$699** — almost the whole principal was already collected. **An early default and a late default look identical to a yes/no classifier but have a ~\$10K swing in profit.** That is exactly why A, B, and the NPV decision all require a model of timing, not just probability.



5. The default trajectory (Deliverable B intuition)

Deliverable B asks, for each origination cohort and loan age, the cumulative fraction of *your approved* loans that have defaulted by that age — a curve that can only rise. Below is the empirical cumulative-default curve over all approved loans; a real submission produces one such monotone curve per cohort (13×13 grid) with 90% bands. It comes directly from aggregating the shared hazard model, which is why B reuses A’s engine.

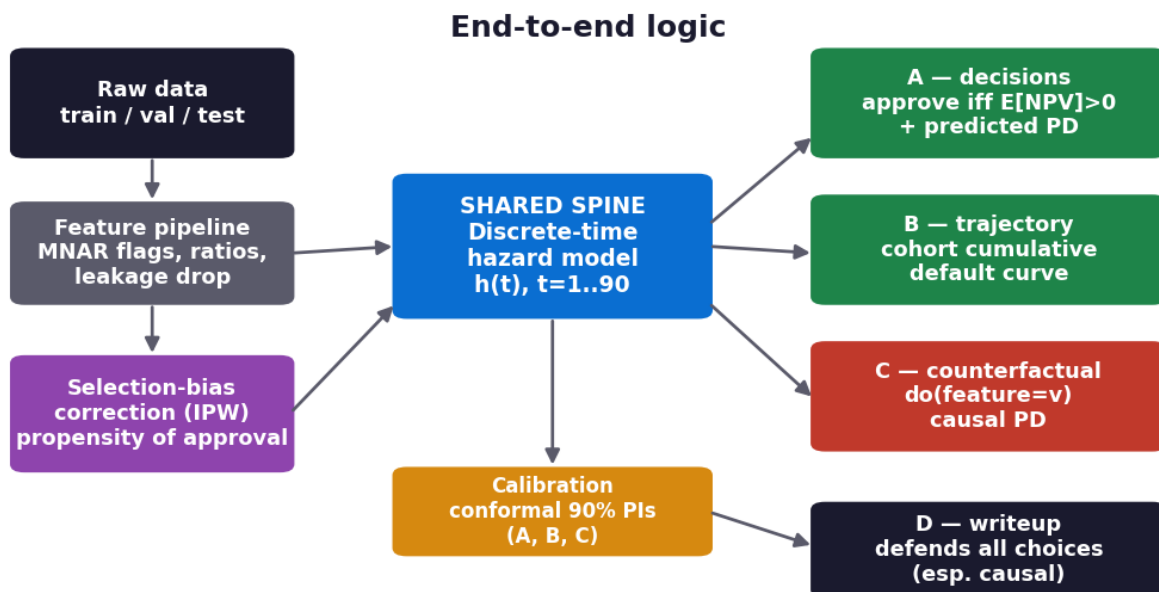
Empirical cumulative default rate vs. loan age (Deliverable B shape)



6. End-to-end logic — one spine, four deliverables

The architecture deliberately centers on a single **discrete-time hazard model** that outputs, per applicant, $h(t) = P(\text{default on day } t \mid \text{survived to } t)$ for $t = 1..90$. Everything else is a read-off of that object:

- **A (decisions)**: survival curve $\rightarrow P(\text{default})$ and the timing distribution $\rightarrow$ expected NPV $\rightarrow$ approve iff $E[\text{NPV}] > 0$. Also emit predicted PD + 90% interval.
- **B (trajectory)**: aggregate the same per-day hazards over the approved loans in each cohort $\rightarrow$ monotone cumulative-default curve + band.
- **C (counterfactual)**: reuse the feature representation but answer $\text{do}(\text{feature} = v)$ with a causal estimator, contrasted against the naive re-prediction.
- **Selection-bias correction (IPW)**: a propensity-of-approval model reweights the labeled loans so the hazard model reflects all applicants, not just previously-approved ones.
- **Calibration**: conformal / quantile methods tune the 90% intervals on held-out validation for $\sim 90\%$ coverage without being needlessly wide (0.20 of the score).
- **D (writeup)**: defends each choice; §3 (causal) carries the most weight.



7. How each deliverable is produced

	Rows	Core method	Main gotcha
A decisions	13,306	hazard -> NPV sign	PD required even for declines; intervals ordered
B trajectory	169	aggregate hazards by cohort	must be monotone non-decreasing in age
C counterfactual	900	causal do(), not re-predict	confounding; ~16 intervenable features only
D writeup	<=4 pg	narrative from A/B/C + EDA	\$3 causal weighted most; 11pt / 0.75in margins

Hard gate: exact file names, flat folder, and `validate_submission.py` must print PASS (it checks IDs, [0,1] ranges, lower $\leq$ point $\leq$ upper, the full 13×13 grid, and B monotonicity). A failing validator is disqualified, so we keep a passing baseline in `out/` at all times.

8. Where we are now (Phase 0 complete)

- Reproducible scaffold in `src/` — data loaders, schema, NPV engine (self-test passes), submission writers, validator harness, EDA + this report generator.
- A complete, **validator-PASS** baseline submission sits in `out/` (A: 13,306, B: 169, C: 900) — we can never be in a non-submittable state.
- Next (Phase 1): feature pipeline (MNAR flags, ratios) + the discrete-time hazard model, compared across families on a common validation NPV metric.

One command reproduces Phase 0: `python run.py phase0`. This document: `python -m src.report_pdf`.